

4 (a) density = mass/volume C1

$$\text{mass} = 7900 \times 4.5 \times 24 \times 10^{-6} = 0.85 \text{ (0.853) kg}$$

M1 [2]

(b) pressure = force/area C1

$$\text{force} = W \cos 40^\circ$$

C1

$$\begin{aligned} \text{pressure} &= (0.85 \times 9.81 \cos 40^\circ) / 24 \times 10^{-4} \\ &= 2.7 \text{ (2.66)} \times 10^3 \text{ Pa} \end{aligned}$$

A1 [3]

(c) $F = ma$ C1

$$W \sin 40^\circ - f = ma$$

C1

$$0.85 \times 9.81 \times \sin 40^\circ - f = 0.85 \times 3.8$$

$$f (= 5.36 - 3.23) = 2.1 \text{ N [5.38 - 3.242 if 0.8532 kg is used for the mass]}$$

A1 [3]

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